

1(a)	scalars: kinetic energy, power, work	A1
	vectors: acceleration, force, momentum	A1
1(b)(i)	mass = volume $\times$ density <i>or</i> $m = V \times \rho$ = $\frac{4}{3} \pi (23 \times 10^{-2})^3 \times 82$	C1
	weight = $\frac{4}{3} \pi (23 \times 10^{-2})^3 \times 82 \times 9.8 = 41 \text{ N}$	A1
1(b)(ii)	vertical component of tension = $290 \sin 75^\circ$ or $290 \cos 15^\circ (= 280)$	C1
	upthrust = $290 \sin 75^\circ + 41$ = 320 (321) N	A1
1(b)(iii)	the water pressure is greater than the air pressure <i>or</i> the pressure on lower surface (of sphere) is greater than the pressure on upper surface (of sphere)	B1